

# **INVISIBLE CLOAK**

A Synopsis

for

Minor Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &  
ENGINEERING**

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## • Introduction

Computer Vision is a field of artificial intelligence that enables computers to interpret and process visual information from images and videos. With the advancement of image processing techniques, real-time video manipulation has become widely used in areas such as augmented reality, gaming, and visual effects.

The proposed project, *Invisible Cloak using Computer Vision*, demonstrates a real-time invisibility effect using Python and OpenCV. The system detects a specific color in a live video stream and replaces that region with a previously captured background frame, creating the illusion that the object or person wearing that color has disappeared.

## • Literature Review

Several computer vision applications rely on color detection and background subtraction techniques to manipulate video streams. Traditional chroma keying methods, commonly used in film production (green screen technology), replace a selected color with a different background.

Research in real-time image segmentation and masking using OpenCV shows that HSV (Hue, Saturation, Value) color space provides better accuracy for color detection compared to RGB. Morphological operations are also widely used to remove noise and improve mask quality.

This project builds upon these established techniques and applies them in a simplified educational implementation using Python and OpenCV.

## • Problem Definition

Real-time object removal or invisibility effects require efficient detection and segmentation of specific regions in a video stream. However, implementing such effects in a simple and accessible manner for educational purposes can be challenging.

The problem addressed in this project is to design and implement a real-time system that detects a predefined color in a live video feed and replaces it with the background to create an invisibility effect using computer vision techniques.

## • Objectives

### Primary Objectives:

- To develop a real-time invisibility cloak system using Python and OpenCV.
- To detect a specific color using HSV color space.
- To replace detected color regions with a static background frame

### Secondary Objectives:

- To understand image segmentation and masking techniques.
- To apply morphological operations for noise reduction.
- To gain hands-on experience in real-time video processing.

## • Methodology

The project follows a systematic approach:

- Capture a static background frame using a webcam.
- Continuously read live video frames from the camera.
- Convert each frame from BGR to HSV color space.
- Define a specific color range and create a mask to detect that color.
- Apply morphological operations to remove noise from the mask.
- Replace the detected region with the stored background frame.
- Display the processed video output in real time.

## References

### For Example:

The references are given for example only. These are not real citations.

### Journal References

[1]. Fuschini F., Falciassecca G., “Experimental investigation of the effects of snowdrifts and ice deposit on linear wire antennas at 2.4 GHz.”, IEEE Antennas and wireless propagation letters, Volume 13, pp: 686-689, 2014.

### Conference References

[2]. Zhigang D., Qahouq J.A.A., “Modelling and investigation of magnetic resonance couple wireless power transfer system with lateral misalignment”, Twenty-Ninth Annual IEEE applied power electronics conference and exposition”, pp: 1317-1322, 2014.

### Books References

[3]. Balanis C.A., “Antenna Theory: Analysis and design”, John Wiley & Sons, Inc., Second edition, pp: 1-4, 1997.

### URL References

[4]. Microstrip patch antenna calculator <http://www.emtalk.com/mpacalc.php>, accessed on 4/8/2022 at 10:30 AM.

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